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# **THERMODYNAMICS OF NONEQUILIBRIUM PHASE TRANSITIONS IN DRIVEN POTTS MODELS**

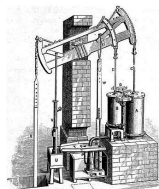
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**Tim Herpich**  
University of Luxembourg

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XVI Latin American Workshop on Nonlinear Phenomena

19th century thermodynamics:

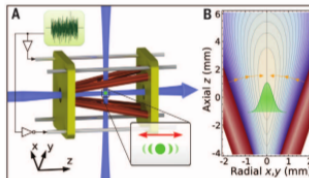
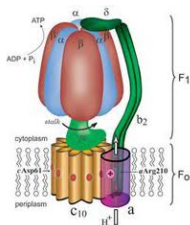


No dynamics

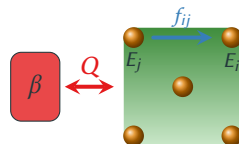
21st century thermodynamics:

Small  $\rightarrow$   $\left\{ \begin{array}{l} \text{Fluctuations} \\ \text{Far-from-equilibrium} \\ \text{Quantum effects} \end{array} \right.$

Build thermodynamics on dynamics



- System with states  $i$  of energy  $E_i$  and occupation probability  $p_i$  interacting with a heat bath at inverse temperature  $\beta$  and subjected to a nonconservative force  $f$ .



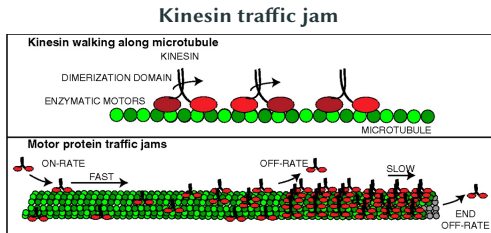
- Markovian master equation and local detailed balance

$$\dot{p}_i = \sum_j W_{ij} p_j, \quad \ln \frac{W_{ij}}{W_{ji}} = -\beta (E_i - E_j - f_{ij})$$

- First law of thermodynamics:  $d_t \langle E \rangle = \langle \dot{W} \rangle + \langle \dot{Q} \rangle$ 
  - Energy current  $d_t \langle E \rangle = d_t \sum_i E_i p_i = \sum_{i,j} (E_i - E_j) W_{ij} p_j$
  - Work current  $\langle \dot{W} \rangle = \sum_{i,j} f_{ij} W_{ij} p_j$
  - Heat current  $\langle \dot{Q} \rangle = \sum_{i,j} (E_i - E_j - f_{ij}) W_{ij} p_j = -\frac{1}{\beta} \sum_{i,j} W_{ij} p_j \ln \frac{W_{ij}}{W_{ji}}$
- Second law of thermodynamics:  $\langle \dot{S} \rangle = d_t \langle S \rangle - \langle \dot{S}_e \rangle \geq 0$ 
  - Nonequilibrium entropy  $\langle S \rangle = -\sum_i p_i \ln p_i$
  - Entropy flow  $\langle \dot{S}_e \rangle = -\sum_{i,j} W_{ij} p_j \ln \frac{W_{ij}}{W_{ji}} = \beta \langle \dot{Q} \rangle$
  - Entropy production  $\langle \dot{S} \rangle = \sum_{i,j} W_{ij} p_j \ln \frac{W_{ij} p_j}{W_{ji} p_i} \geq 0$
- Detailed balance at equilibrium  $f_{ij} = 0$ :  $W_{ij} p_j^{eq} = W_{ji} p_i^{eq} \Rightarrow \langle \dot{S} \rangle = 0$

- **Nonequilibrium phase transitions** are most commonly discussed in the framework of nonlinear dynamics and bifurcation theory
- **Stochastic thermodynamics** provides a systematic treatment of nonequilibrium thermodynamics  $\Rightarrow$  **thermodynamics of nonequilibrium phase transitions**
- **Practical motivation:** Increasing power by considering ensembles of nanomachines
  - Can we tune interactions to improve performance?
  - How do phase transitions affect the performance?
  - Is there an optimal number of interacting nanomachines?

- **Example:**

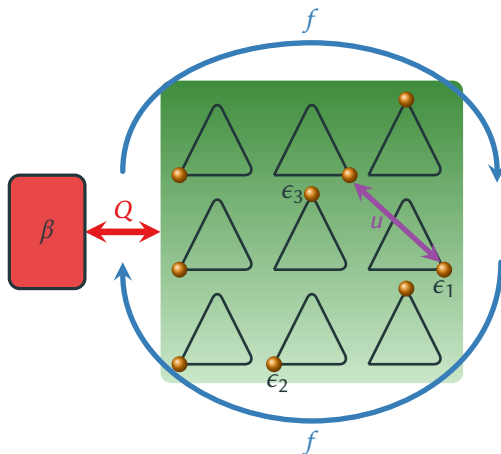


- **Synchronization** seems like a promising candidate to improve performance

$\Rightarrow$  We look for a thermodynamically consistent model for synchronization

# DRIVEN THREE-STATE POTTS MODEL

- **Network** of  $N$  all-to-all coupled three-state [PRL **96**, 145701 (2006)] units.
- System in contact with **heat bath**.
- State:  $\mathbf{N} = (N_1, N_2, N_3)$ .
- **All-to-all interactions** (pair energy  $u$ )
- **Interaction energy**  $U(\mathbf{N}) = \frac{u}{2N} \sum_{i=1}^3 N_i(N_i - 1)$
- Subjected to **torque**  $f$  acting as a global bias



## Stochastic Dynamics

- Markovian master equation with Arrhenius rates satisfying local detailed balance

$$\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N},\mathbf{N}'} P_{\mathbf{N}'}, \quad \ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}}} = -\beta [\Delta F^{eq}(\mathbf{N}, \mathbf{N}') - \Theta(\mathbf{N}, \mathbf{N}') f], \quad F^{eq}(\mathbf{N}) = E(\mathbf{N}) - \beta^{-1} S^{int}(\mathbf{N})$$

## Thermodynamics

### First law

$$d_t \langle E \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} E(\mathbf{N}) W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'} = \langle \dot{Q} \rangle + \langle \dot{W} \rangle$$

$$\langle \dot{Q} \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} [E(\mathbf{N}) - f \Theta(\mathbf{N}, \mathbf{N}')] W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'}$$

$$\langle \dot{W} \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} f \Theta(\mathbf{N}, \mathbf{N}') W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'}$$

### Second law

$$d_t \langle S \rangle \equiv d_t \sum_{\mathbf{N}} P_{\mathbf{N}} [S^{int}(\mathbf{N}) - \ln P_{\mathbf{N}}] = \langle \dot{S}_e \rangle + \langle \dot{S} \rangle$$

$$\langle \dot{S}_e \rangle \equiv - \sum_{\mathbf{N}, \mathbf{N}'} W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'} \left[ \ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}}} + \Delta S^{int}(\mathbf{N}, \mathbf{N}') \right] = \beta \langle \dot{Q} \rangle$$

$$\langle \dot{S} \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'} \ln \frac{W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}} P_{\mathbf{N}}} \geq 0$$

## Stochastic Dynamics

- Mean-field approximation is exact for  $N \rightarrow \infty$  and yields **2D nonlinear master equation**

$$\dot{\bar{n}}_i(t) \equiv \lim_{N \rightarrow \infty} \langle N_i(t)/N \rangle = \sum_{j=1}^3 k_{ij} \bar{n}_j(t), \quad \sum_{i=1}^3 \bar{n}_i(t) = 1$$

with mean-field transition rates satisfying **local detailed balance**

$$\ln \frac{k_{ij}}{k_{ji}} = -\beta [\epsilon_i - \epsilon_j + u(\bar{n}_i - \bar{n}_j) - \Theta(i, j)f]$$

## Thermodynamics

### First law

$$d_t \langle \mathcal{E} \rangle \equiv \sum_{i,j} \mathcal{E}_i k_{ij} \bar{n}_j = \langle \dot{Q} \rangle + \langle \dot{W} \rangle$$

$$\langle \dot{Q} \rangle \equiv \sum_{i,j} [\mathcal{E}_i - f \Theta(i, j)] k_{ij} \bar{n}_j$$

$$\langle \dot{W} \rangle \equiv \sum_{i,j} f \Theta(i, j) k_{ij} \bar{n}_j$$

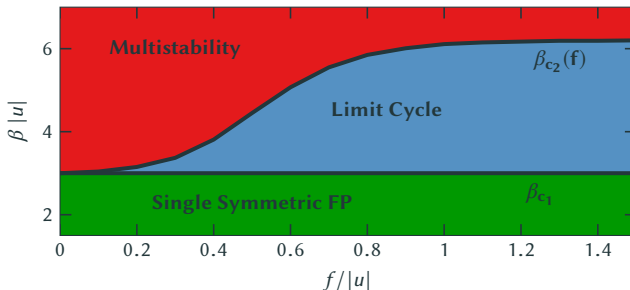
### Second law

$$d_t \langle \mathcal{S} \rangle \equiv -d_t \sum_i \bar{n}_i \ln \bar{n}_i = \langle \dot{S}_e \rangle + \langle \dot{S}_i \rangle$$

$$\langle \dot{S}_e \rangle \equiv - \sum_{i,j} k_{ij} \bar{n}_j \ln \frac{k_{ij}}{k_{ji}} = \beta \langle \dot{Q} \rangle$$

$$\langle \dot{S}_i \rangle \equiv \sum_{i,j} k_{ij} \bar{n}_i \ln \frac{k_{ij} \bar{n}_j}{k_{ji} \bar{n}_i} \geq 0$$

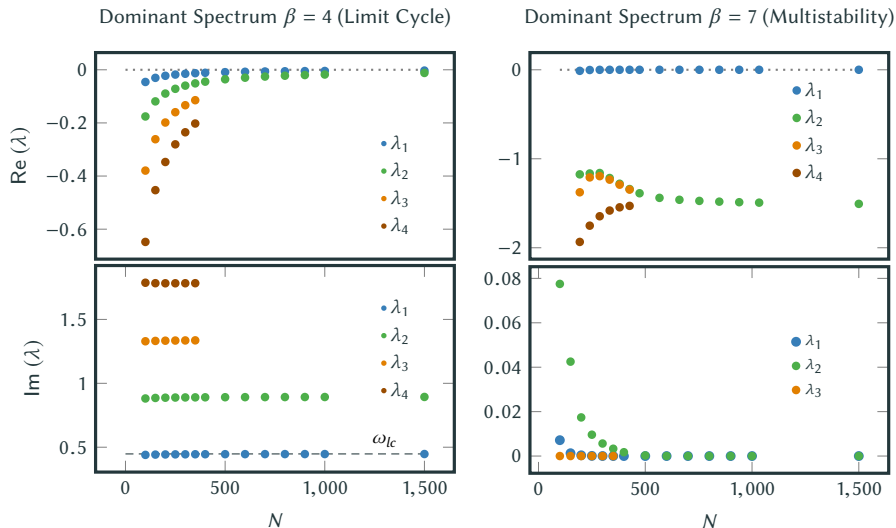
- For  $\epsilon_i = \epsilon$ , the nonlinear equation has a **symmetric fixed point**  $\dot{\bar{n}}_i = \sum_{j=1}^3 k_{ij} \bar{n}_j |_{\bar{n}_j=1/3} = 0$
- Linear stability indicates **Hopf bifurcation**:  $\lambda_{\pm} = -\Gamma(\beta u + 3) \cosh\left(\frac{\beta f}{2}\right) \pm i\sqrt{3}\Gamma \sinh\left(\frac{\beta f}{2}\right)$



Herpich, Thingna, and Esposito, PRX 8, 031056 (2018)

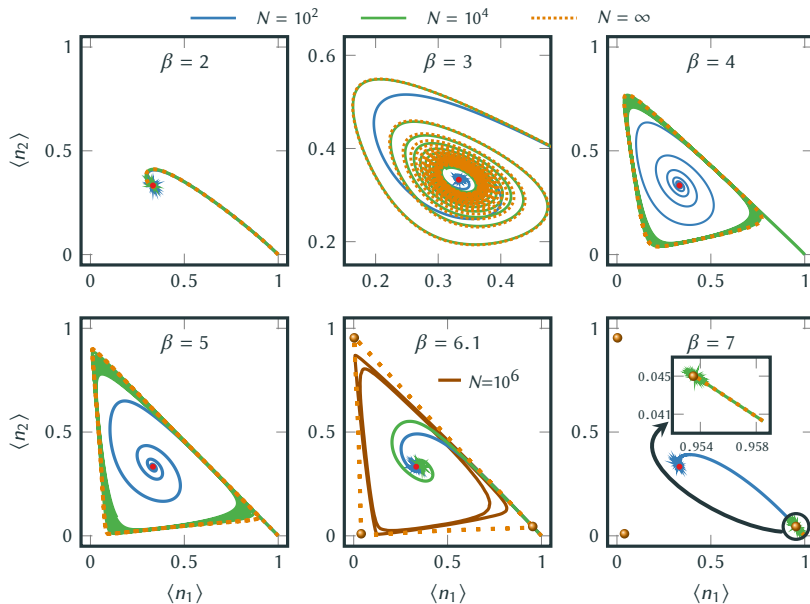
- One equilibrium (first-order) and two nonequilibrium phase transitions at  $\beta = \beta_{c1,2}$
- Three distinct fixed points in the **low-temperature regime** ( $\beta \gg \beta_c$ ) Energy-driven
- Symmetric solution in the **high-temperature regime** ( $\beta \ll \beta_c$ ) Entropy-driven
- Thermodynamic consistency constraints **stable oscillations** to emerge only at intermediate temperatures ( $\beta_{c1} < \beta \leq \beta_{c2}$ )

- Apparent paradox of how **linear irreducible Markov dynamics** can converge to **nonlinear mean-field dynamics** that exhibits complex solutions  $\rightarrow$  **Metastability?**

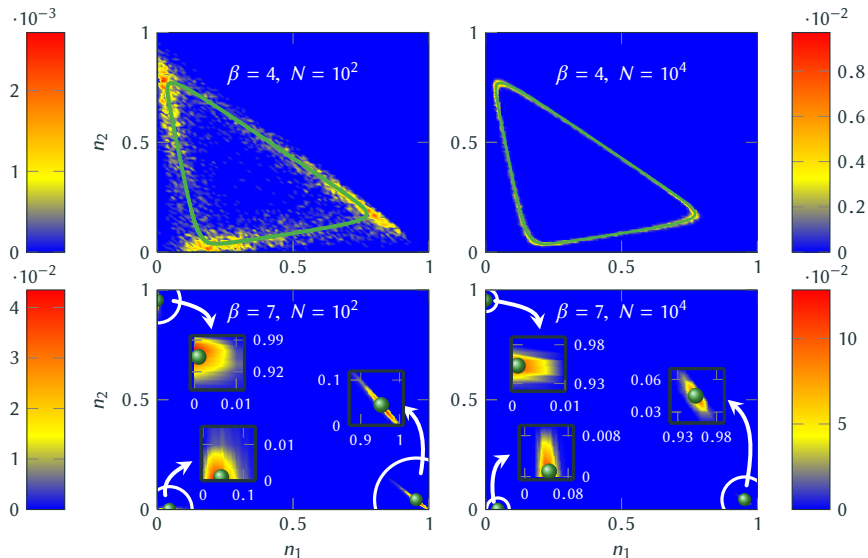


- Slowest finite-time mode** encodes the **complex mean-field solutions** and persists over increasing time scales as  $N$  grows.

- Finite  $N$  vs. mean field:



Is metastability a generic property of our system or merely an artifact of special initial conditions?



Herpich, Thingna, and Esposito, PRX 8, 031056 (2018)

$$N = 10^2$$

$$\beta = 4$$

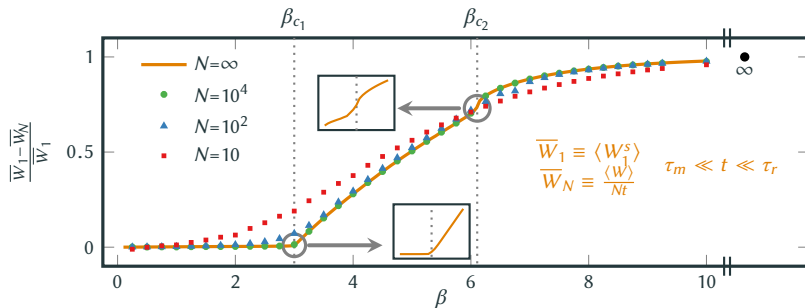
$$N = 10^4$$

$$N = 10^2$$

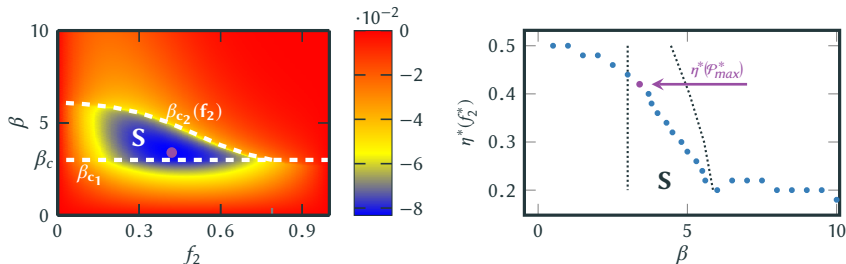
$$\beta = 7$$

$$N = 10^4$$

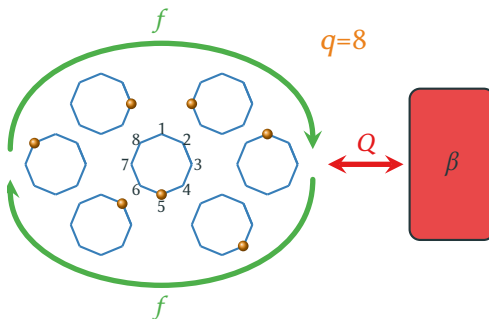
- Relative Reduction in Dissipated Work due to Interactions:



- Output Power and Efficiency at Maximum Power ( $f = f_1 - f_2$ , Fuel  $f_1 = 1$ , Output  $f_2$ ):



- Generalize the setup to driven  $q$ -state units on a ring.



- Mean-field dynamics is governed by  $(q - 1)$ -dimensional nonlinear master equation

$$\dot{\bar{n}}_i(t) = \sum_{j=1}^q k_{ij}(\bar{n}_i, \bar{n}_j) \bar{n}_j(t), \quad \ln \frac{k_{ij}}{k_{ji}} = -\beta [\epsilon_i - \epsilon_j + u(\bar{n}_i - \bar{n}_j) - \sigma(i, j)f]$$

- Flat energy landscape  $\epsilon_i = \epsilon \forall i$  and attractive interactions  $u < 0$ :
- Symmetric solution  $\bar{n}_i = 1/q$  in the **high-temperature limit**
- $q$  energy-ground states  $\bar{n}_i = \delta_{ij}$  in the **low-temperature limit**
- Intermediate** temperatures?

Entropy-driven

Energy-driven

- Linear stability analysis of  $\bar{n}_i^* \equiv 1/q$  yields for **the critical point**  $q + \beta_c u = 0$  for  $q \leq 4$
- The linear stability matrix**  $\mathbf{A}_{ij} \equiv \sum_k [\partial(k_{ik} \bar{n}_k) / \partial \bar{n}_j] |_{\bar{n}^*}$  evaluated at  $\beta_c = -u/q$  reads

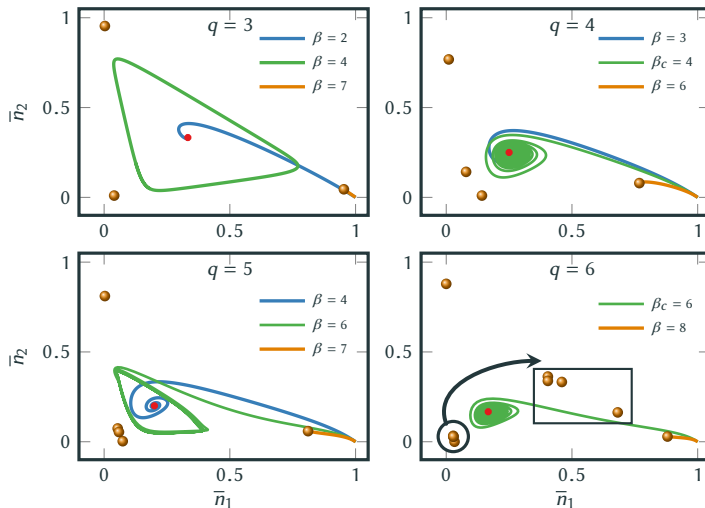
$$\mathbf{A} = \begin{pmatrix} 0 & c & 0 & \dots & 0 & 0 & -c \\ -c & 0 & c & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -c & 0 & c \\ c & 0 & 0 & \dots & 0 & -c & 0 \end{pmatrix}_{q \times q} \quad \begin{aligned} c &= \Gamma \sinh \left( q \frac{f}{2u} \right) \\ \lambda_k &= 2i c \sin \left( k \frac{2\pi}{q} \right), \quad k = 0, 1, \dots, q-1 \end{aligned}$$

- Perturbation** of  $\mathbf{A}(\beta_c + \delta\beta) = \mathbf{A}(\beta) |_{\beta_c} + \delta\beta \mathbf{B} + \mathcal{O}(\delta\beta^2)$ , yields

$$\mathbf{B} = \begin{pmatrix} a & b & 0 & \dots & 0 & 0 & c \\ c & a & b & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & c & a & b \\ b & 0 & 0 & \dots & 0 & c & a \end{pmatrix}_{q \times q} \quad \begin{aligned} a &= -\frac{2\Gamma u}{q} d, \quad b = \Gamma \frac{2u - qf}{2q} d, \\ c &= \Gamma \frac{qf + 2u}{2q} d, \quad d = \cosh \left( q \frac{f}{2u} \right), \\ \tilde{\lambda}_k &= -\frac{2\Gamma d \sin^2 \left( \frac{k\pi}{q} \right)}{q} \left[ 2u + ifq \cot \left( \frac{k\pi}{q} \right) \right], \quad k = 0, 1, \dots, q-1 \end{aligned}$$

- In absence of driving,  $f = 0$ , the real eigenvalues change sign  $\rightarrow$  saddle-node bifurcation
- In presence of driving,  $f \neq 0$ , the real part of the eigenvalues change sign  $\rightarrow$  **Hopf bifurcation**

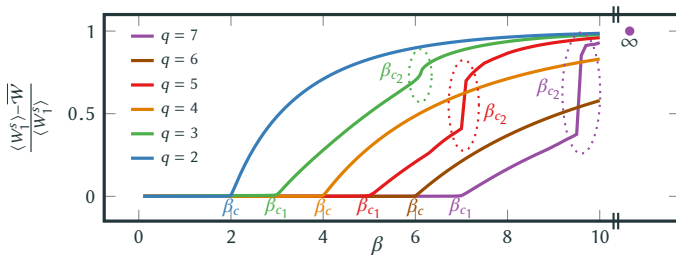
- Stability of oscillations?



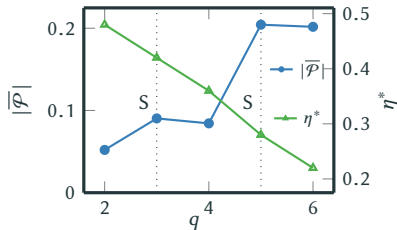
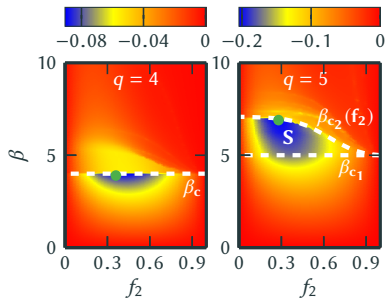
- If  $q$  is **even** there is only a **subcritical Hopf bifurcation**, if  $q$  is **odd** there are two bifurcations: a **supercritical Hopf** and an **infinite-period bifurcation**

# THERMODYNAMICS OF THE POTTS MODEL

- Dissipated work



- Power-efficiency trade-off



- Summary

- Two classes of Potts models for even or odd  $q$  with one or two nonequilibrium phase transitions
- Reduction of the dissipated work for any finite interactions
- The maximum power is achieved for a synchronous operating mode and the associated efficiency of this far-from-equilibrium scenario is close to linear-response value
- $\Rightarrow$  Small and odd- $q$  Potts models in synchronization regime are energetically most beneficial

- Perspectives

- Local interactions
- Fluctuations across nonequilibrium phase transitions
- Further studies on impact of collective effects for machine performance

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Thank you for your attention!